Adaptive-subtraction Electromagnetic Interference Noise Reduction in Photoacoustic Imaging in Vivo

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***Abstract*—Photoacoustic imaging (PAI) has garnered substantial interest due to its unique ability to provide high-resolution, high-contrast visualization of biological tissues. Nevertheless, electromagnetic interference (EMI) remains a considerable impediment, compromising the fidelity of the acquired images. This study delineates a methodology devised to mitigate electromagnetic noise within the PAI system. By employing advanced signal processing algorithms, specifically using sorting and median filtering techniques, we effectively attenuated EMI, thereby enhancing the peak signal-to-noise ratio (PSNR) of the detected photoacoustic signals. The efficacy of the proposed approach was validated through a series of controlled experimental evaluations, utilizing both phantom models and in vivo assessments. The resultant data corroborate a marked reduction in noise levels, with a concomitant improvement in image clarity and contrast. These findings substantiate the potential of our approach to significantly elevate the precision and diagnostic utility of PAI in clinical and research applications.**

Keywords—Photoacoustic imaging, electromagnetic interference, signal processing, , noise reduction.

I. Introduction

Photoacoustic imaging (PAI) is an advanced biomedical imaging technique that combines the high spatial resolution of optical imaging with the deep tissue penetration of ultrasound [1]. This hybrid approach allows for the non-invasive visualization of anatomical and functional information at the molecular level, making it particularly valuable in medical diagnostics and research [2].

In photoacoustic tomography (PAT), the use of impulse lasers is essential for generating high-resolution images [3]. However, this reliance on impulse lasers makes the system particularly susceptible to electromagnetic interference (EMI) noise if adequate electromagnetic shielding is not implemented [4]. While electromagnetic shielding can effectively mitigate this issue [5], it presents significant challenges. Not only is the process of electromagnetic shielding costly, but it also adds considerable complexity to the system's design and implementation.

Significant efforts have been devoted to mitigating EMI noise in PAI through advancements in deep learning. Notably, recent research has focused on leveraging deep learning techniques to enhance the separation of electromagnetic noise from the photoacoustic signal. Christoph Dehner et al. [6] proposed a UNet-based deep learning denoising method that leverages spatiotemporal correlations within the entire photoacoustic sinogram. The model is trained on a large dataset combining experimentally acquired noise and synthetic signals. Similarly, Oleksandra Gulenko et al. [7] developed a deep learning approach tailored to distinguish and eliminate electromagnetic noise in photoacoustic endoscopy.

While these deep learning-based methods have shown remarkable performance in specific applications, they are not without significant limitations [8]. One of the primary challenges is the reduction in robustness when there is a variation in test samples, indicating that the models may not perform consistently across different datasets [9]. Additionally, these methods exhibit a high dependency on large, annotated datasets, which are often difficult and time-consuming to acquire in practice. Moreover, the generalization capabilities of these models are frequently constrained, meaning that their effectiveness may be limited when applied to scenarios or data distributions that differ from those used during training [10]. These limitations highlight the inherent challenges of deep learning approaches in photoacoustic imaging, reinforcing the value and potential advantages of traditional methods, which may offer greater robustness and generalization without the heavy reliance on large datasets [11].

In this work, we explored the feasibility of traditional methodologies for isolating EMI noise from photoacoustic signals. We devised an algorithm called Adaptive Electromagnetic Interference Subtraction (AEIS), predicated on sorting and median filtering, to attenuate EMI noise, which adeptly identifies and excises temporal segments corrupted by electromagnetic noise. Validation was conducted using simulated data, wherein background noise was synthesized and superimposed onto photoacoustic sinograms generated via the k-Wave toolbox [12]. Subsequently, we applied the developed algorithm to clinical data procured from human finger imaging. The algorithm exhibited superior performance in terms of peak signal-to-noise ratio (PSNR), underscoring its efficacy and potential for practical applications in clinical environments.

II. Methods

In this part, we detail the AEIS algorithm for removing EMI noise from Photoacoustic sinograms, as shown in Fig. 1.

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Additionally, we reconceptualize the denoising problem as a decomposition task. Leveraging this reformulation, we design a discriminative algorithm tailored for electromagnetic noise reduction.

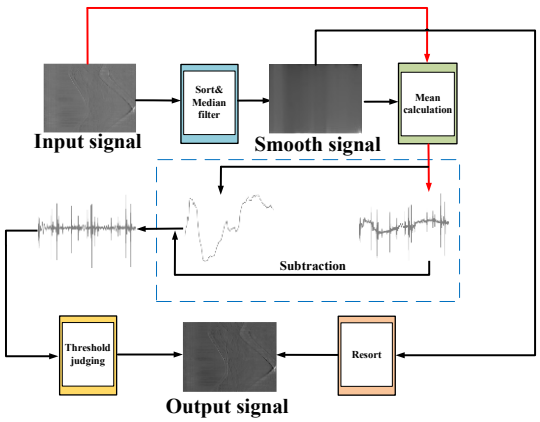


Fig. 1. The flow chart of proposed EMI noise removal method.

A. *Denoising via Decomposition*

Let p be the observed photoacoustic sinogram of $M * N$ which suffers from EMI noise, and s is the clean sinogram. Mathematically, the photoacoustic sinogram degradation models can be formulated as:

$$p = s + n \tag{1}$$

Where n is the electromagnetic noise caused by parasitic noise and thermal noise [4]. While the thermal noise is easy to model as Gaussian noise. But the parasitic noise is shown in Fig. 2. as a impulse noise which is hard to model, and because its frequency spectrum significantly overlaps with the photoacoustic signal, traditional frequency domain filtering cannot be used to eliminate these noises.

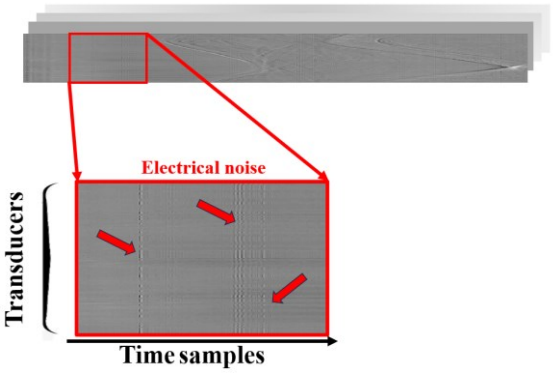


Fig. 2. Electrical noise in photoacoustic sinogram.

B. *Detailed Examination of the AEIS Denoising Algorithm*

Each column of the sinogram is first sorted based on signal values, after which a median filter is applied to the sorted image $S_{\text{smooth}} \in \mathbb{R}^{M\times N}$.

We calculate the mean of each column of S_{smooth}, , resulting in a column mean vector:

$$\mu_j = \frac{1}{M}\sum_{i=1}^{M} S_{\text{smooth}}(i,j),\ j = 1,2,\dots,N \tag{2}$$

Step 2: Compute the Overall Mean of Pre-Sorted Columns and Subtract. Before performing the sorting operation, we compute the overall mean of all column means:

$$\mu_{\text{global}} = \frac{1}{N}\sum_{j=1}^{N} \mu_j \tag{3}$$

Next, we compute the difference between each column mean and the overall mean:

$$\Delta\mu_j = \mu_j - \mu_{\text{global}},\ j = 1,2,\dots,N \tag{4}$$

Step 3: Compare with Adaptive Threshold to Detect EMI Noise.

Here we take the threshold $T_{\text{adaptiver}}$ as the average value before time. The method for obtaining this threshold, as tested by us, effectively addresses the threshold issue. We then compare the difference with the threshold to detect the presence of EMI noise: if $|\Delta\mu_j| > T_{\text{adaptive}}$, then EMI noise is detected in column.

Step 4: Reorder Columns to Obtain the Corrected Sinogram. For the columns where EMI noise is detected, we reorder them back to their original rows to obtain the corrected sinogram:

$$S_{\text{corrected}} = \text{Reorder}(S_{\text{smooth}}) \tag{5}$$

Here, Reorder $(\cdot)$ denotes the operation of reordering the columns back to their original row positions.

III. Experiments and Results

To evaluate the effectiveness of the proposed algorithm, we conducted experiments using actual human finger data. The photoacoustic signals were generated by a Nd laser (CNI Inc., China) with a wavelength of 532 nm, a pulse energy of 20 mJ, and a repetition rate of 10 Hz. For signal detection, a 128-element circular ultrasonic transducer with a center frequency of 2.5 MHz and a bandwidth of 60% was employed.

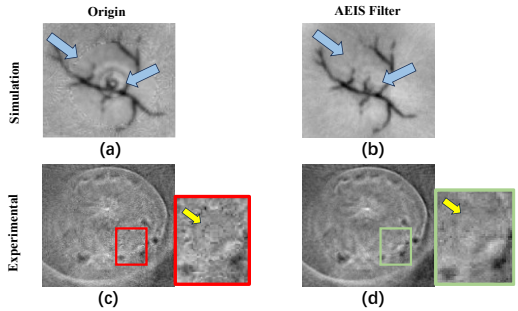


Fig. 3. The reconstruction results using DAS. (a) Simulated image without noise reduction. (b) Simulated image after AEIS algorithm. (c) Experimental image of a human finger without noise reduction. (d) Experimental image after AEIS algorithm.

The results shown in Fig. 3. demonstrate that the AEIS algorithm efficiently eliminates EMI noise and notably enhances the PSNR. For both simulation and in vivo human finger experiments, we observed a significant enhancement in imaging contrast upon removing EMI noise, whose quantitative evaluation is summarized in Table 1.

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Table. 1. Signal and Image Quality Metrics.

	Signal Quality		Image Quality	
Type	**SNR (dB)**	**PSNR (dB)**	**SSIM**	**PSNR (dB)**
Origin (Signal)	-13.101	8.792	-	-
AEIS Filter (Signal)	18.04	26.42	-	-
Origin (Image)	-	-	0.549	11.873
AEIS Filter (Image)	-	-	0.721	13.302

IV. DISCUSSION AND CONCLUSION

In this work, we propose an electromagnetic noise reduction algorithm designed to mitigate the noise caused by the laser system. We conducted experiments on both simulated data and live human finger data, demonstrating that our approach effectively eliminates the majority of electromagnetic noise.

For future work, we plan to explore several avenues to further enhance the effectiveness and applicability of our electromagnetic noise reduction algorithm. One potential direction is to refine the algorithm to handle a broader range of noise sources, including those with more complex patterns or varying intensities. Another key area for future research is the integration of our algorithm with other noise reduction techniques to improve overall robustness. Finally, we intend to validate our approach on a wider variety of biological tissues and under different experimental conditions to assess its generalizability and reliability in diverse clinical settings.

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